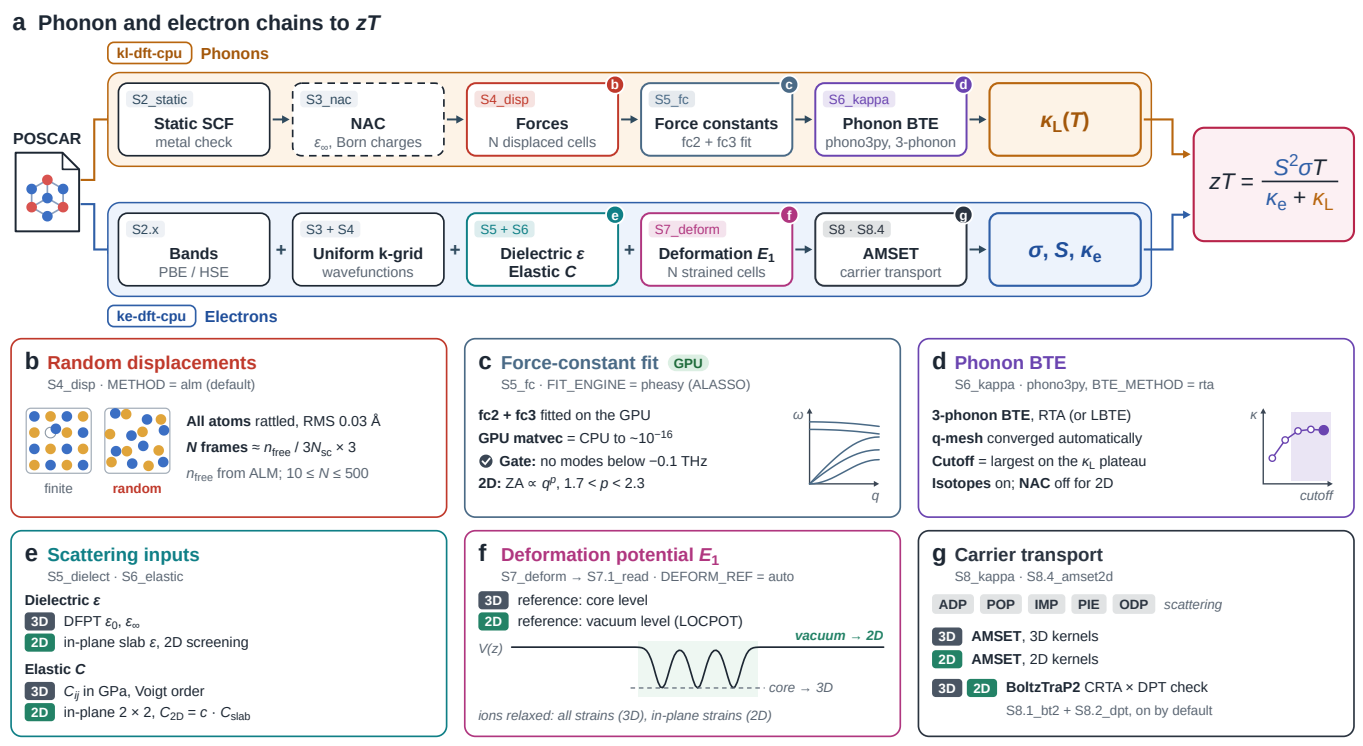

a Phonon and electron chains to zT
kl-dft-cpu
Phonons
POSCAR
S2_static
Static SCF
metal check
S3_nac
NAC
ε_∞, Born charges
S4_disp
b
Forces
N displaced cells
S5_fc
c
Force constants
fc2 + fc3 fit
S6_kappa
d
Phonon BTE
phono3py, 3-phonon
$\kappa_L(T)$
S2.x
Bands
PBE / HSE
S3 + S4
Uniform k-grid
wavefunctions
S5 + S6
e
Dielectric ε
Elastic C
S7_deform
f
Deformation E_1
N strained cells
S8 · S8.4
g
AMSET
carrier transport
σ, S, κ_e
$zT = \frac{S^2\sigma T}{\kappa_e + \kappa_L}$
ke-dft-cpu
Electrons
b Random displacements
S4_disp · METHOD = alm (default)
finite
random
All atoms rattled, RMS 0.03 Å
N frames ≈ n_{free} / $3N_{sc}$ × 3
n_{free} from ALM; 10 ≤ N ≤ 500
c Force-constant fit
GPU
S5_fc · FIT_ENGINE = pheasy (ALASSO)
fc2 + fc3 fitted on the GPU
GPU matvec = CPU to ~10^{-16}
Gate: no modes below −0.1 THz
2D: ZA ∝ q^p, 1.7 < p < 2.3
ω
q
d Phonon BTE
S6_kappa · phono3py, BTE_METHOD = rta
3-phonon BTE, RTA (or LBTE)
q-mesh converged automatically
Cutoff = largest on the κ_L plateau
Isotopes on; NAC off for 2D
κ
cutoff
e Scattering inputs
S5_dielect · S6_elastic
Dielectric ε
3D DFPT ε_0, ε_∞
2D in-plane slab ε, 2D screening
Elastic C
3D C_{ij} in GPa, Voigt order
2D in-plane 2 × 2, $C_{2D} = c \cdot C_{slab}$
f Deformation potential E_1
S7_deform → S7.1_read · DEFORM_REF = auto
3D reference: core level
2D reference: vacuum level (LOCPOT)
V(z)
vacuum → 2D
core → 3D
ions relaxed: all strains (3D), in-plane strains (2D)
g Carrier transport
S8_kappa · S8.4_amset2d
ADP POP IMP PIE ODP scattering
3D AMSET, 3D kernels
2D AMSET, 2D kernels
3D 2D BoltzTraP2 CRTA × DPT check
S8.1_bt2 + S8.2_dpt, on by default